Knowing When Not to Answer

Abstention Drift in Multi-Turn Retrieval-Augmented Conversations

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GitHub Repository: `https://github.com/SK-099/abstention-drift`

Deployed Project: `https://abstain-rag.duckdns.org`

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Table 1: Project group.

1 Introduction

Chat assistants that answer from a user's own documents use retrieval-augmented generation (RAG) [4]: they retrieve passages and let a language model answer from them. Their most harmful failure is answering confidently when the documents do *not* contain the answer. The goal of this project was to build and deploy a complete, self-hosted notebook assistant that (i) answers only from the user's sources, with citations, (ii) says "I don't know" when the sources do not support an answer, also late in a long conversation, and (iii) runs on modest hardware. The GitHub repository ships all ten supported models, and the complete application runs locally with Docker at `http://localhost:3000`; because of limited disk space and memory on the AWS virtual machine, only five of these ten models were deployed there. A research stage measured abstention on the MTRAG-UN benchmark and chose the refusal method; an engineering stage turned it into the deployed application.

2 Related Work

SQuAD 2.0 [5] made unanswerable questions a standard test, and RGB [6] showed that RAG models often answer from irrelevant passages. Surveys of abstention [7] and multi-model approaches [8] study when a model should decline; selective prediction [9, 10] chooses refusal thresholds under a risk budget, the approach we use for every model. Self-RAG [11] checks its own answers, similar to our self-check step. MTRAG [1] and MTRAG-UN [2] provide multi-turn RAG conversations with unanswerable questions; at SemEval-2026 a strong system still detected only 21.8% of them on the development set [3]. Existing source-grounded notebook assistants are mostly closed, cloud-only services; our application provides this workflow with open models that run locally.

3 Problem Definition

Given a notebook of sources (PDFs, web pages, YouTube transcripts, pasted text) and a conversation, the assistant must, at every turn, either give an answer whose statements are supported by cited passages, or refuse. The requirements for the deployed system were:

- **Grounded, cited answers** and a measured, low rate of both wrong answers to unanswerable questions (*abstain recall*) and wrong refusals of answerable ones (*false-abstention rate*, FA).
- **Choice of model**: ten free open-weight models (1–4B parameters, via Ollama), each with its own refusal thresholds, since models differ strongly in how often they refuse on their own.
- **Notebook features**: source guides, notebook-wide summaries, Studio outputs (briefing, study guide, FAQ, timeline, mind map, flashcards, quiz, two-voice audio overview), notes and sharing.
- **Deployability**: one-command start with Docker Compose, locally with all ten models (`http://localhost:3000`) and on an AWS EC2 instance with five of them, with automated tests on every change.

4 Experiments and Results

4.1 System built

The assistant is a LangGraph agent behind a FastAPI service (Figure 1). For each question it rewrites the question using the chat history, runs hybrid search (BM25 [12] and BGE embeddings [13] fused with reciprocal rank fusion [14], then a cross-encoder re-ranker [15]) over the notebook's passages in Qdrant, and scores answerability with a fine-tuned checker (gate). It abstains if the gate score $p \geq \tau$, if the model itself refuses, or if $p \geq$ min_p and a self-check finds the answer unsupported; before refusing it searches once more with a rewritten query. A Celery worker indexes sources and generates Studio outputs; Kokoro text-to-speech voices the audio overview.

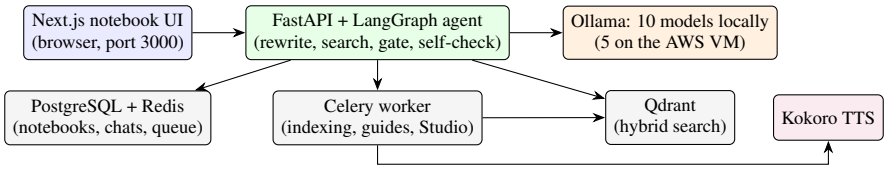


Figure 1: Architecture (Docker containers; Ollama on the host). Locally all 10 models in the repository are available; the AWS VM, limited in disk and memory, runs 5 of them.

4.2 Choosing the refusal method

On the MTRAG-UN test set (429 turns, 97 unanswerable) we compared a fixed pipeline with answerability gates and with the agent (Table 2). Detection of unanswerable questions did not get worse in later turns (all slopes near zero, confidence intervals contain zero), so no turn-specific fix was needed. The main weakness was *over-refusal*; the agent's "search again before refusing" step lowered it to 13.6%, and adding the calibrated self-check gave the best F1, so this configuration was deployed.

Configuration (Qwen2.5-3B)	F1	Abstain recall	Precision	FA ↓	Recall change per turn [95% CI]
Hybrid retrieval, prompt-only refusal	.443	.464	.425	.184	−.003 [−.046, +.043]
+ fine-tuned answerability gate	.449	.495	.410	.208	+.008 [−.034, +.051]
Agent: retry before refusing	.465	.443	**.489**	**.136**	+.010 [−.033, +.054]
Agent: retry + self-check (deployed)	**.511**	**.577**	.459	.199	+.020 [−.026, +.068]

Table 2: Abstention on MTRAG-UN (429 turns).

4.3 Per-model refusal thresholds

For each of the ten models we chose $(\tau, \min_p)$ on 224 MTRAG development turns to maximise F1 while keeping FA at most 10 points above the model's own refusal rate. The best models reached F1 .67 (Gemma-3-4B, Qwen3-1.7B) against .62 for the research model Qwen2.5-3B; the best thresholds ranged from $\tau = .45$ to .64, so one setting cannot serve all models.

4.4 End-to-end testing

A test script drove the running application through its API with three research papers and all ten models (Table 3); continuous integration on GitHub additionally runs lint, 88 unit tests, the web build and an abstention quality gate on every push.

Feature tested (all 10 models)	Result	Status
Chat: answerable questions	48/60 answered with citations to the right sections (experiments questions cite Experiments/Results)	pass
Chat: unanswerable questions	26/30 refused; only DeepSeek-R1-1.5B and SmolLM2-1.7B (own refusal rate .09/.02) answered	pass
Sources	PDF, web page, pasted text, YouTube (timestamped citations), web search; source guides	pass
Studio	briefing, study guide, FAQ, timeline, mind map, flashcards, quiz; audio overviews of 60–230 s	pass
Sharing and notes	viewer/editor roles, link access and revoking tested with separate users; notes stay private	pass
Overall	184 checks in the main run, 0 failures; 22 warnings were model behaviour, not faults	pass

Table 3: End-to-end test results.

4.5 Deployment

	Laptop	AWS EC2 (free-plan credits)
Hardware	16 GB RAM, NVIDIA GTX 1650 (4 GB)	`m7i-flex.large`: 2 vCPU, 8 GB RAM, 40 GB disk, no GPU
Models	all 10 models in the repository	5 of the 10, chosen to fit the limited disk and memory
Typical answer time	10–70 s	tens of seconds (CPU only)
Access	`http://localhost:3000` (Docker)	`https://abstain-rag.duckdns.org`
Cost	–	about USD 0.10 per hour; stopped when not in use

Table 4: Deployment environments.

Problems found and fixed during deployment. Ollama's default 4,096-token window silently cut the sources out of long Studio prompts, so small models wrote about unrelated topics (fixed with a larger window and an off-topic check); reference lists were retrieved as evidence (filtered); and containers each using all CPU threads slowed answers about tenfold (thread limits).

5 Conclusion

We built and deployed a self-hosted notebook assistant that answers from the user's sources with citations and refuses when they do not support an answer. Measurements on MTRAG-UN guided the design: abstention does not drift over a conversation, the real problem is over-refusal, and an agent that searches again and checks its answer gives the best balance (F1 .511). Per-model thresholds make ten open models usable; all ten pass the end-to-end test when run locally with Docker, while the AWS instance serves five of them. Remaining limitations are slow CPU-only answers in the cloud, no sign-in for public use, and a gate that is less reliable on scientific papers; calibrating it on users' own documents and adding authentication are the next steps.

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